

Muhammad Umar

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Objective

3rd-year AI undergraduate at GIKI who builds and ships real software. Deployed a **facial recognition attendance system** used daily by faculty, a **real-time campus food delivery platform**, and a **biomedical knowledge graph** with 2.1M+ triples. Experienced across the full stack — from training deep learning models to deploying production APIs on Linux servers. Proficient in Python, JavaScript, TensorFlow, FastAPI, and PostgreSQL.

Education

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology Topi, Khyber Pakhtunkhwa, Pakistan

Bachelor of Science in Artificial Intelligence

2023 – 2027

- Relevant Coursework: Software Engineering · DBMS · Advanced Linear Algebra · Inferential Statistics · Knowledge Graphs · DSA · Machine Learning

Projects

Smart Attendance System | Python · YOLOv9 · ArcFace · FastAPI · React Native · PostgreSQL · Docker

- **Deployed on GIKI's internal Linux server and actively used by faculty** for real-time attendance marking via mobile app.
- **YOLOv9 + ArcFace** facial recognition pipeline achieving **99%+ accuracy** across varying lighting and angles.
- FastAPI backend with JWT auth, OTP verification, and role-based access control; containerized with **Docker** on a Unix-based host.
- PostgreSQL database for student records, attendance logs, course enrollments, and user credentials.

GIKI Panda – Campus Food Delivery System | Python · Flask · PostgreSQL · Socket.IO · Leaflet.js

- Multi-role platform (Student, Vendor, Rider, Admin) with **real-time GPS tracking** via WebSockets.
- Designed RESTful API endpoints with clear separation of concerns between front-end and back-end layers.
- Anti-spoofing layer: rate limiting, teleport detection, and campus geofencing for security.
- Live Demo: giki-panda-food-2f1e85dc6df5.herokuapp.com

Biomedical Knowledge Graph System | Python · OWL 2 · SPARQL · GraphDB · Flask · D3.js

- Constructed a **2.1M+ triple** biomedical knowledge graph from PubMed using OWL 2 ontology with 23 classes and HermiT reasoning.
- Python ETL pipeline processing **50K+ records** with automated entity linking to DBpedia and Wikidata.
- Flask + D3.js visualization dashboard with live SPARQL endpoint, deployed on Heroku.
(krr-685beba13d3f.herokuapp.com)

Experience

Research Intern – AI Model Development for Epilepsy Seizure Prediction

June 2025 – July 2025

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)

- Built personalized **CNN-BiLSTM with attention** for patient-specific seizure prediction achieving **75%+ accuracy** using EEG, MRI, and clinical records.
- Designed a real-time prediction pipeline in **TensorFlow** for large-scale neurophysiological data processing.
- Processed multimodal data using TMSST-based signal processing and collaborated with a medical research team to translate clinical requirements into working software.

Skills

Languages: Python, JavaScript, C++, SQL

ML & AI: TensorFlow, PyTorch, Scikit-learn, YOLOv9, ArcFace, CNN-BiLSTM, OpenCV, Pandas, NumPy

Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets, JWT Authentication

Databases: PostgreSQL, MySQL, MongoDB, GraphDB, SPARQL, OWL 2

Tools & Platforms: Docker, Git, GitHub, Heroku, Linux (Ubuntu), Postman

Certifications

Advanced Google Data Analytics – Coursera

Full Stack Development – Scrimba